Finite Length and Multiscale Boundary Selection in Burg Mirror Descent

Abstract

We prove a general-clock finite-length theorem for inexact Burg descent on a product of positive free coordinates and a simplex. For positive steps tending to zero with divergent accumulated time T, actual capture of the free block and ordinary dual residual $O(\beta T^{-3/2})$ imply convergence to a unique convex analytic minimum, an inverse-clock objective bound, finite physical length, and exact literal boundary profiles. No square-summability or regularity of successive steps is required. The proof combines inherited cone-adapted static geometry with a residual-aware exact discrete clock argument. We verify its hypotheses for nonlinear positive-concave perspectives: a unique secondary minimum is selected on a harmonic clock; a fully flat secondary face with positive transverse quadratic form yields a unique tertiary minimum on any face, including zero gaps and degenerate supported curvature. Exact reciprocal order supplies a complementary nonunique support criterion. An explicit finite-depth family has sharp iterated-logarithmic physical tails, separating rate-free length from uniform tail claims. All perspective reductions retain the original normalized map and ambient step restriction.

Keywords: Burg mirror descent; variable steps; boundary selection; finite physical length; nonlinear perspectives; logarithmic clocks

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1 The exact map and the hierarchy of clocks

We study boundary selection and finite physical length for Burg descent. The organizing result is an inexact variable-step theorem on a product of positive free coordinates and a simplex. It separates the actual accumulated time from the iteration index, and is then applied to two exact normalizations of nonlinear perspective objectives. A reciprocal-order criterion and a sharp logarithmic cascade describe complementary nonunique behavior and delimit what a general tail estimate could mean.

Write

$$\Delta_d = \{x \in \mathbb{R}^d : x_i \geq 0,\ \mathbf{1}^T x = 1\}, \qquad h(x) = -\sum_i \log x_i \quad (x \in \operatorname{ri} \Delta_d).$$

The original physical map used by all perspective applications is

$$X_{k+1} = \operatorname*{arg\,min}_{z \in \operatorname{ri} \Delta_d} \{\eta \langle \nabla F(X_k), z - X_k \rangle + D_h(z, X_k)\}, \qquad X_0 \in \operatorname{ri} \Delta_d, \quad 0 < \eta L < 1, \tag{M.1}$$

with the dimension understood from the coordinates in each application. Here $D_h(z, x) = h(z) - h(x) - \langle \nabla h(x), z - x \rangle$ and L is an admitted ambient Euclidean Hessian bound on

the original simplex. No smaller chart-dependent step restriction is substituted. Its exact reciprocal identity is

$$\Delta(1/X_i) = \eta\partial_i F(X_k) + \lambda_k, \qquad \lambda_k = -\eta\frac{\sum_i X_{i,k}X_{i,k+1}\partial_i F(X_k)}{\sum_i X_{i,k}X_{i,k+1}}. \tag{M.2}$$

The multiplier is the actual endpoint-product weighted mean, not a limiting KKT multiplier. All lengths refer to sums of physical Euclidean step norms; finite-dimensional ℓ^1 length is equivalent. We use $\Delta X_k = X_{k+1} - X_k$ and $\mathcal{L}_k = \sum_{l\geq k}\|\Delta X_l\|_2$.

What is proved and what is reused. Section 2 states the general-clock engine. Its dynamic inputs are explicit: positive vanishing steps with divergent sum, actual capture of the positive free block, and a quantitative ordinary dual residual. Section 3 verifies these inputs for nonlinear perspectives first on a harmonic clock and then, under controlled transverse geometry, on a logarithmic clock of that harmonic clock. Neither reduction replaces transient normal variables by their equilibrium values. Section 4 gives a separate elementary rate-free support criterion, and Section 4.1 gives exact tails at every finite logarithmic depth.

The finite cone-spanned flag and literal source-metric algebra are inherited static machinery, reproduced with full proofs in Supplement Section S1. They do not themselves prove a variable-step convergence theorem. The new clock proof in Supplement Section S2 establishes entry, the residual-aware gap, reciprocal capture, the actual scaled map, lifted descent, physical length and profile selection in its own clock. The application proof in Supplement Section S4 separately establishes both exact reductions and their error budgets. This distinguishes reusable geometry from the dynamical assertions depending on it.

1.1 Relation to established convergence mechanisms

The convex Burg comparator supplies the established $O(\log k/k)$ value bound for the original map [4, Theorem 2, equation (20)]. Diminishing-step mirror-descent iterate convergence is also established: Doan, Bose, Nguyen and Beck [10, Theorem 1] consider nonincreasing positive square-summable, nonsummable steps and a Lipschitz convex objective, using finite comparator Bregman divergences. A forced-zero Burg optimizer is outside the barrier domain. Our proof approaches such optimizers only after interior comparison, allows arbitrary positive vanishing divergent steps, and additionally controls physical length and literal boundary profiles.

Interior-cluster Bregman length theorems [3, Theorems 17–19] do not directly cover a forced-zero Burg limit. Analyticity alone also does not supply the usual Euclidean relative-error hypothesis [5, Section 2.3 and Theorem 2.9] when normal mobility vanishes quadratically. The scaled-KL framework has separate applicability conditions: its Burg

half-line example fails its extended scaled-KL property, whereas its automatic subanalytic result assumes a closed kernel domain [2, Proposition 4]. Its Example 5 also separates bounded physical Burg flow from unbounded logarithmically transformed flow. These are obstructions to direct use of that proof framework, not nonconvergence examples. We instead apply the ordinary analytic gradient inequality to a compact lifted set and project the resulting estimate onto the physical path.

Vanishing steps and inexact variable metrics are already treated by Frankel, Garrigos and Peypouquet [13, Theorem 1, H1–H3]. A concrete obstruction to direct application on the original Burg objective is $f(s, 1-s) = s$. Its constrained interior Euclidean slope is a positive constant, while the exact reciprocal difference

$$\Delta \left[\frac{1}{s} - \frac{1}{1-s} \right] = \beta_k > 0$$

makes s_k decrease and gives finite physical total variation. Their relative-error hypothesis H2 would require $b_{k+1} \operatorname{dist}(0, \partial f(X_{k+1})) \leq \|\Delta X_k\| + \varepsilon_{k+1}$. With summable additive errors, summing forces $\sum b_k < \infty$, contrary to H3. This rules out that direct application to the original objective, not a new lifted application. Constructing the captured lift and its physical projection estimates is therefore essential here.

Azizian, Iutzeler, Malick and Mertikopoulos [14, Definition 1, Assumption 2, Theorem 1 and Remark 3] already connect boundary geometry and sharpness to rates, including cumulative-step formulations for variable steps. Their stated regularizer is finite on the solution domain, and their second-order sufficiency condition imposes quadratic growth in the relevant operator sense. A forced-zero Burg solution violates that finite-domain condition, while higher-order supported degeneracy need not meet the growth assumption.

For the reparameterization theorem of Ding and Toh [9, Definition 3.1 and Theorem 4.1], the Burg metric-flattening coordinate is $\log x_i$ up to a constant. A vanishing physical coordinate therefore violates both compact transformed closure and the stated bounded-transformed-orbit alternative. Physical finite length can coexist with infinite logarithmic-coordinate length. Their variable-step framework also allows positive nonsummable diminishing steps; Lemma 4.12 does not require a positive lower step bound. Our distinction is therefore the infinite-distance Burg boundary, physical projected length, controlled residual and capture, and exact profiles, rather than the removal of a lower step bound alone.

Convex finite-type weighted decompositions are classical; see Schulz [7] and the formulation in Kamimoto [8, Lemma 2.1]. An unrestricted rotation need not preserve literal missing coordinates. The cone-spanned version reproduced here retains those rows and their exact reciprocal witnesses. The real-analytic gradient inequality [1, 6] is a standard input. General mirror-descent convergence cannot be inferred for arbitrary kernels: Bolte and Pauwels [11, Corollaries 10 and 12] construct nonconvergence for a specially chosen Legendre kernel and a linear objective. Their kernel is not the fixed separable Burg kernel

considered here.

Iterated logarithmic hierarchies in optimization are already present, for example, in the gradient-descent expansion of Soudry et al. [12, Appendix C, Theorem 13]. The cascade below concerns a different exact discrete map, compact-simplex boundary selection and a sharp finite physical length tail. Barrier-induced variational selection also has established examples. For a linear objective with zero dual error, exact Burg descent samples the cumulative barrier path; Alvarez, Bolte and Brahic [15, Propositions 3.2 and 4.4] already give the relevant path and optimal-face selection structure. Linear strict-gap constants are recovered behavior here. Full-space finite-type flags and least-common-multiple weighted dilations are also inherited geometric structure, not a contribution claim. The nonlinear exact-map residual closure, physical projection and verified secondary/tertiary reductions require the additional arguments supplied below.

2 A general-clock boundary engine

Repeated normalization changes the effective time of a Burg orbit. The following theorem works with the actual accumulated step, and permits a controlled error in the dual equation. Its positive free coordinates are useful for normal blow-ups; their capture is an assumption about the actual sequence, not a consequence of a formal equilibrium equation. The theorem does not include a nonvanishing fixed effective step. Original fixed-step perspective orbits will instead produce actual vanishing-step lifts.

Theorem 1 (Inexact Burg descent on an arbitrary divergent clock)**.** *Let $\mathcal{P} = (0,\infty)^r \times \Delta_d$, $r \geq 0$, $d \geq 1$, and let f be convex and real analytic on an open convex neighborhood of $\mathcal{P}$. Suppose it has a unique minimizer $w_* = (a_*, s_*)$ in $\mathcal{P}$. For the positive free block it suffices to have this extension on a neighborhood of a compact box containing the orbit, a_*, and their convex hull, times the closed simplex. Put*

$$h(a,s) = -\sum_j \log a_j - \sum_i \log s_i.$$

Let an actual interior sequence $w_k = (a_k, s_k)$ satisfy

$$\nabla h(w_{k+1}) - \nabla h(w_k) = -\beta_k \nabla f(w_k) + (0, \lambda_k \mathbf{1}) + e_k, \tag{M.3}$$

where λ_k is its actual common simplex multiplier. Assume

$$\beta_k > 0, \quad \beta_k \to 0, \quad \sum_k \beta_k = \infty, \qquad 0 < c \leq a_{j,k} \leq C, \tag{M.4}$$

$$T_k = 1 + \sum_{l=k_0}^{k-1} \beta_l, \qquad \|e_k\| \leq C\beta_k T_k^{-3/2}. \tag{M.5}$$

Then $w_k \to w_*$,

$$f(w_k) - f(w_*) = O(T_k^{-1}), \qquad \sum_k \|w_{k+1} - w_k\|_2 < \infty.$$

All boundary KKT gap patterns and supported analytic degeneracies are allowed. There is a fixed cone-adapted affine chart $w = w_* + J\theta$, *an even degree* ℓ, *and a diagonal integer-weight dilation* $D(\rho)$ *such that, for* $\rho_k = T_k^{-1/\ell}$,

$$\theta_k = D(\rho_k) z_k, \qquad z_k \to z_*.$$

Write the leading objective and literal missing rows as

$$f(w_* + JD(\rho)z) - f(w_*) = \rho^\ell [Q(z) + O(\rho)], \qquad s_i = \rho^{b_i} l_i(\rho, z).$$

Then z_* *uniquely minimizes*

$$V_0(z) = Q(z) - \frac{1}{\ell} \sum_{i : s_{i,*} = 0} b_i \log l_i(0, z) \quad \text{on } l_i(0, z) > 0. \tag{M.6}$$

In particular,

$$T_k^{b_i/\ell} s_{i,k} \longrightarrow l_i(0, z_*) > 0, \qquad T_k[f(w_k) - f(w_*)] \longrightarrow \frac{1}{\ell^2} \sum_i b_i^2. \tag{M.7}$$

A positive-gap row has $b_i = \ell$ *and limit* $1/\gamma_i$, *where* γ_i *is its constrained KKT gap; each such row contributes one to the energy limit. Zero-gap missing rows have* $0 < b_i/\ell \le 1/2$. *Empty sums are zero.*

The complete proof is Supplement Section S2. Only the static cone/flag and literal Gram algebra of Sections S1.1, S1.3, S1.4 and S1.2 are reused, after checking their product-domain hypotheses. Entry, the residual-aware inverse-clock gap, capture, the changed-clock exact equation, lifted descent and profile selection are proved there for (M.3). Neither square summability of β_k, nor monotonicity, ratios or differences of successive steps are assumed. The exponent $3/2$ is a uniform sufficient residual budget, not an optimality assertion. Superalgebraic and exponentially small errors in T satisfy it. The length conclusion concerns w, not unweighted variation of the further blow-up z.

Dynamic profile versus ordinary barrier profile. The weight-dependent logarithmic coefficient in (M.6) is a dynamical dilation correction. For comparison, the ordinary barrier objective $f + T^{-1}h$, with the same parameter T, gives the leading rescaled profile $Q - \sum_i \log l_i(0, z)$ in this chart; protected logarithms contribute only a constant at leading order. At its profile minimum, weighted Euler gives $Q = \sum_i b_i/\ell$. The endogenous dynamic weights b_i/ℓ instead give $\sum_i (b_i/\ell)^2$. They agree for literal positive gaps, where $b_i = \ell$, but need not agree for degenerate rows. This compares the two profile functionals only; no general central-path convergence or capture theorem is used. The calculation distinguishes the nonlinear profiles; it is not a claim that barrier-path methods cannot inform the analysis. Supported components of z_* can vanish, so no nonzero leading equivalent is asserted for every supported coordinate.

2.1 A cone-essential example with mixed literal rows

The following single example exhibits two flag degrees, a full-space zero direction removed by feasibility, and a singular mixed literal Gram. It illustrates Theorem 1 on its vanishing-step clock, rather than invoking a fixed-step theorem.

On Δ_5 with coordinates (a, b, z, w, u) let

$$F = (a+b)^2 + (z-2w)^2 + (z+w)^4. \tag{M.8}$$

This entire convex polynomial has the unique simplex minimum $p_* = (0, 0, 0, 0, 1)$, with four zero KKT gaps. Write $d = z-2w$ and $s = z+w$. The degree-two term is $(a+b)^2+d^2$. Its full kernel contains the line $a = -b, d = s = 0$, on which F vanishes identically. The feasible-cone intersection removes this line: it forces $a = b = d = 0$, leaving only $s \geq 0$. The next degree is four. Thus the cone-adapted degree is $\ell = 4$, with weights $(2, 2, 2, 1)$ on (a, b, d, s).

For $\rho = T^{-1/4}$ put $(a, b, d, s) = (\rho^2 A, \rho^2 B, \rho^2 D, \rho S)$. Then

$$F = \rho^4[(A+B)^2 + D^2 + S^4], \qquad z = \frac{\rho}{3}(2S + \rho D), \qquad w = \frac{\rho}{3}(S - \rho D). \tag{M.9}$$

The literal row bases are $(2, 2, 1, 1)$, and the profile is

$$\begin{aligned} V_0 = (A+B)^2 + D^2 + S^4 - \tfrac{1}{2}\log A - \tfrac{1}{2}\log B \\ - \tfrac{1}{4}\log(2S/3) - \tfrac{1}{4}\log(S/3). \end{aligned} \tag{M.10}$$

Its unique minimum is $A_* = B_* = 1/\sqrt{8}$, $D_* = 0$, $S_* = 8^{-1/4}$. Every interior-start exact Burg orbit with $\beta_k > 0$, $\beta_k \to 0$ and $\sum \beta_k = \infty$ therefore has finite physical length and, with $T_k = 1 + \sum_{j<k} \beta_j$,

$$\begin{aligned} &T_k^{1/2} a_k,\ T_k^{1/2} b_k \longrightarrow 1/\sqrt{8}, \qquad T_k^{1/2}(z_k - 2w_k) \longrightarrow 0, \\ &T_k z_k^4 \longrightarrow 2/81, \qquad T_k w_k^4 \longrightarrow 1/648, \qquad T_k F_k \longrightarrow 5/8. \end{aligned} \tag{M.11}$$

These are accumulated-clock limits, not k-power laws for arbitrary schedules.

The two logarithmic rows in (D, S) lose rank as $\rho \to 0$; at $D = 0$ their Gram block is

$$\begin{pmatrix} 5\rho^2/(4S^2) & -\rho/(2S^2) \\ -\rho/(2S^2) & 2/S^2 \end{pmatrix}.$$

Thus diagonalizing the objective does not produce independent scalar Burg coordinates: D is signed, there is no $\log D$, and the literal z, w logarithms mix both directions. A positive slower S makes either sign of the faster D feasible. The gradient has no globally minimizing coordinate, so the global common-anchor criterion does not explain the example. Supplement Section S3 verifies well-definedness, all engine hypotheses, the flag, metric, constants and failure of global order explicitly. This does not rule out a special-purpose proof or an eventual orbit-dependent anchor.

3 Nonaffine perspectives on successive clocks

The general-clock theorem applies to positive concave analytic denominators after the required free-coordinate capture and dual residual have been established on the original orbit. A unique secondary minimum needs one normalization. A flat secondary face with a uniformly positive transverse quadratic form admits a second normalization and a tertiary selection problem.

3.1 A unique secondary minimum

On the unit simplex in coordinates $X = (n, x)$, with $q \geq 1$ normal coordinates and $d \geq 1$ tangential coordinates, consider

$$F(n,x) = \sum_{j=1}^{q} \frac{n_j^{p_j}}{A_j(x)^{p_j-1}}, \qquad p_j \geq 2 \text{ even}. \tag{M.12}$$

Each A_j is positive, concave and real analytic on an open convex neighborhood of the full projected simplex $\{x \geq 0 : \mathbf{1}^T x \leq 1\}$. Use the exact map (M.1), an interior start, and its original ambient step $0 < \eta L < 1$, with L an ambient Euclidean Hessian bound for F. Define

$$\Phi(s) = -\sum_j \frac{p_j - 1}{\eta p_j^2} \log A_j(s), \qquad s \in \Delta_d, \tag{M.13}$$

$$V(Z,s) = \sum_j \frac{Z_j^{p_j}}{A_j(s)^{p_j-1}} - \sum_j \frac{1}{\eta p_j} \log Z_j, \qquad Z \in (0,\infty)^q. \tag{M.14}$$

For a fixed sufficiently late $k_0 \geq 1$, put

$$\begin{gathered} Z_{j,k} = k^{1/p_j} n_{j,k}, \qquad t_k = 1 - \sum_j n_{j,k}, \qquad s_k = x_k/t_k, \\ \beta_k = \eta/k, \qquad \tau_k = 1 + \sum_{l=k_0}^{k-1} \beta_l \sim \eta \log k, \qquad w_k = (Z_k, s_k). \end{gathered} \tag{M.15}$$

Theorem 2 (Unique secondary selection on the harmonic clock)**.** *Suppose Φ has a unique minimizer s_* on Δ_d. It may lie on any face, with arbitrary KKT gaps and supported analytic degeneracy. Then every orbit above has finite Euclidean and ℓ^1 length in its original coordinates. Moreover the augmented sequence w_k itself has finite length and*

$$\begin{gathered} w_k \longrightarrow w_* = (Z_*, s_*), \qquad Z_{j,*}^{p_j} = \frac{A_j(s_*)^{p_j-1}}{\eta p_j^2}, \\ V(w_k) - V(w_*) = O(\tau_k^{-1}) = O((\log k)^{-1}). \end{gathered} \tag{M.16}$$

All the flag and profile conclusions of Theorem 1 apply with $f = V$ and $T = \tau$. In particular, for its adapted degree ℓ, literal weights b_i and profile minimizer z_*,

$$\tau_k^{b_i/\ell} x_{i,k} \longrightarrow l_i(0, z_*) > 0 \quad (s_{i,*} = 0), \tag{M.17}$$

$$\tau_k[V(w_k) - V(w_*)] \longrightarrow \frac{1}{\ell^2} \sum_{i:s_{i,*}=0} b_i^2.$$

For a missing coordinate with positive constrained secondary gap γ_i of Φ,

$$(\log k) x_{i,k} \longrightarrow \frac{1}{\eta \gamma_i}.$$

Zero-gap missing coordinates have rational exponents $0 < b_i/\ell \le 1/2$. The limit is independent of initialization; no common ordering of the denominator derivatives is assumed.

The exact first lift has residual $O(k^{-1-\epsilon} \log k)$, where $\epsilon = \min_j 1/p_j > 0$. Its ratio to β_k is smaller than every inverse power of τ_k. Supplement Section S4 proves this estimate and the actual positive Z capture before applying Theorem 1.

3.2 A controlled unique tertiary minimum

Now write the original simplex coordinates as $X = (n, u, y)$, with $q, m \ge 1$ normal and exterior coordinates and $d \ge 1$ face coordinates. Retain (M.12), with $x = (u, y)$, all its analytic, concavity and positivity assumptions on the full projected simplex, and the same original ambient step. Assume the full projected flatness identities

$$A_j(u, 0) = a_j > 0, \qquad \partial_y A_j(u, 0) = 0 \quad \text{for every } u \ge 0,\ \mathbf{1}^T u \le 1. \tag{M.18}$$

These are analytic identities on a neighborhood of that slice, including its radial directions. Define

$$H_j(u) = -\partial_{yy}^2 A_j(u, 0), \qquad c_j = \frac{p_j - 1}{\eta p_j^2 a_j}, \qquad H(U) = \sum_j c_j H_j(U), \tag{M.19}$$

and assume

$$H(U) \succeq h_0 I_m \quad (U \in \Delta_d), \qquad h_0 > 0. \tag{M.20}$$

Individual H_j need not be positive definite. Set

$$\begin{gathered} W(R, U) = \frac{1}{2} R^T H(U) R - \frac{1}{2} \sum_{a=1}^{m} \log R_a, \qquad R \in (0, \infty)^m, \quad U \in \Delta_d, \\ \widehat{R}(U) = \arg\min_{R>0} W(R, U), \qquad K_{\mathrm{ter}}(U) = W(\widehat{R}(U), U). \end{gathered} \tag{M.21}$$

The proof establishes that W is jointly convex analytic, $\widehat{R}$ is unique, positive and analytic, and K_{ter} is convex analytic. In addition to Z, t, β, τ in (M.15), use the actual orbit variables

$$\begin{gathered} (b_k, \widetilde{y}_k) = (u_k, y_k)/t_k, \qquad Y_k = \mathbf{1}^T \widetilde{y}_k, \qquad U_k = b_k/(1 - Y_k), \\ R_k = \sqrt{\tau_k}\, \widetilde{y}_k, \qquad \alpha_k = \beta_k/\tau_k, \qquad \sigma_k = 1 + \sum_{l=k_0}^{k-1} \alpha_l = \log \tau_k + O(1) \sim \log\log k. \end{gathered} \tag{M.22}$$

Theorem 3 (Controlled tertiary selection on any face)**.** *Assume* (M.18)–(M.20)*, and suppose* K_{ter} *has a unique minimizer* U_* *on* Δ_d*. No interiority, positive supported Hessian, positive missing-coordinate gap or derivative-order condition is required. Every original orbit has finite Euclidean and* ℓ^1 *physical length and converges to* $(0, U_*, 0)$*, independently of initialization. More precisely,*

$$Z_{j,k} \longrightarrow \left(\frac{a_j^{p_j-1}}{\eta p_j^2}\right)^{1/p_j}, \qquad (R_k, U_k) \longrightarrow (\widehat{R}(U_*), U_*), \tag{M.23}$$
$$\sum_k \|(R_{k+1}, U_{k+1}) - (R_k, U_k)\|_2 < \infty,$$
$$W(R_k, U_k) - \min W = O(\sigma_k^{-1}) = O((\log\log k)^{-1}), \qquad (\log k) y_{a,k}^2 \longrightarrow \frac{\widehat{R}_a(U_*)^2}{\eta}.$$

The complete flag and profile conclusions of Theorem 1 hold for $f = W$ *and* $T = \sigma$*. Thus, with that theorem's adapted degree, literal weights and profile minimizer,*

$$\sigma_k^{b_i/\ell} u_{i,k} \longrightarrow l_i(0, z_*) > 0 \quad (U_{i,*} = 0), \tag{M.24}$$
$$\sigma_k[W(R_k, U_k) - \min W] \longrightarrow \frac{1}{\ell^2} \sum_{i:U_{i,*}=0} b_i^2.$$

In particular a positive constrained tertiary gap γ_i *of* K_{ter} *gives*

$$(\log\log k) u_{i,k} \longrightarrow \frac{1}{\gamma_i}. \tag{M.25}$$

Zero-gap missing coordinates have rational exponents $0 < b_i/\ell \le 1/2$*, even when the supported Hessian is degenerate.*

The secondary minimum set here is the entire face $\widetilde{y} = 0$, generally nonunique. We do not apply Theorem 2 to the first lift. Instead, a moving-face convex comparison and the exact normalization multiplier give $|Z - Z_*| + Y = O(\tau^{-1/2})$ and $c \le R_a \le C$ before any face coordinate is protected. The second exact dual equation has residual $O(\alpha_k \tau_k^{-1/2} \log(e\tau_k))$; divided by α_k, this is exponentially small up to a polynomial factor in its own clock σ. The proof in Supplement Section S4 retains both Z and R as actual dynamical variables.

Neither theorem asserts a universal physical tail-length equivalent. The tertiary statement requires full projected flatness, uniform transverse positive definiteness and a unique K_{ter} minimum. It does not cover proper secondary slices, singular transverse quadratic forms or nonunique tertiary minima.

3.3 The verified hypothesis-transfer interface

Proposition 3.1 (Two demonstrated transfers)**.** For the original exact perspective orbits in Theorem 2 or Theorem 3, respectively, set

$$(a, s, f, \delta, \mathcal{T}) = (Z, s, V, \beta, \tau) \quad \text{or} \quad (R, U, W, \alpha, \sigma).$$

Each chosen reduction verifies the following inputs to Theorem 1 on the actual sequence:

1. The positive free block a is bounded above and away from zero.

2. With its Burg kernel, the dual update is exactly $\Delta\nabla h = -\delta\nabla f + (0, \lambda\mathbf{1}) + e$.

3. Its own clock is $\mathcal{T}_k = 1 + \sum_{l<k} \delta_l$ (with the fixed initial index understood), $\delta_k \to 0$, $\sum \delta_k = \infty$, and $\|e_k\|/\delta_k \le C_N \mathcal{T}_k^{-N}$ for every fixed $N > 0$ on a tail.

4. The reduced objective is jointly convex analytic with a unique minimum, under the corresponding theorem's selection hypothesis.

5. The engine's finite variation reconstructs finite physical variation through the proved normal variation and bounded finite-variation mass factors.

Proof. For the first transfer, Supplement Sections S4.2–S4.3 prove actual capture, the exact identity and its residual, and the minimum/reconstruction statements. For the second, Sections S4.4–S4.8 prove the moving-face and exterior capture estimates, the exact second identity and its own-clock budget, and reconstruction. In the first case $x = ts$; in the second $(u, y) = (1 - N)((1 - Y)U, \tau^{-1/2}R)$. These are exact identities with the stated variation bounds. ∎

This proposition records two verified transfers. It does not assert that an arbitrary nonlinear minimizing face admits another such reduction: the second transfer uses full projected flatness and positive transverse quadratic geometry.

3.4 Two scope examples

A nonordered unique secondary minimum with quartic supported growth. On $n + u + v + z = 1$, take

$$A(u, v, z) = 3 - (u - v)^4 - z^2, \qquad F = n^2/A.$$

The denominator is positive, concave and analytic on a convex neighborhood of the full projected simplex, where $A \ge 1$. Its unique maximum on Δ_3 is $(1/2, 1/2, 0)$, so Theorem 2 applies. The supported $u - v$ direction has quartic leading growth and zero Hessian. Moreover

$$A_u = -4(u - v)^3, \qquad A_v = 4(u - v)^3, \qquad A_z = -2z.$$

The largest denominator derivative changes between u and v; no single coordinate dominates globally. The theorem gives finite physical and augmented length without such an order condition.

A boundary tertiary minimum with quartic supported growth. On $n+u+v+z+y=1$, take

$$B(u,v,z)=3-(u-v)^4-z^2, \qquad A(u,v,z,y)=3-\frac{y^2}{B(u,v,z)}, \qquad F=n^2/A. \tag{M.26}$$

Here $B \geq 1$ and $A \geq 2$ on the full projected simplex. The quadratic-over-positive-denominator composition makes A concave analytic on a convex neighborhood. Full projected flatness holds with $a=3$, and

$$H(U)=\frac{1}{6\eta B(U)}, \qquad \widehat{R}(U)^2=3\eta B(U), \qquad K_{\mathrm{ter}}(U)=\text{constant}-\frac{1}{4}\log B(U).$$

Its unique minimizer is $(U_u,U_v,U_z)=(1/2,1/2,0)$. The missing z direction has zero gap and quadratic growth, while the supported $u-v$ direction has quartic growth. Theorem 3 gives finite physical and tertiary length and

$$kn_k^2 \longrightarrow \frac{3}{4\eta}, \qquad (\log k)y_k^2 \longrightarrow 9, \qquad (\log\log k)z_k^2 \longrightarrow 3. \tag{M.27}$$

Indeed, in local coordinates $q=R-3\sqrt{\eta}$, $d=U_u-U_v$ and $z=U_z$, the degree-four weighted jet, with weights $2,1,2$, is

$$Q(q,d,z)=\frac{q^2}{18\eta}+\frac{d^4+z^2}{12}.$$

Its sole missing divisor has weight two. The profile is $Q-\frac{1}{2}\log z$, whose minimum has $q=d=0$ and $z^2=3$. It also yields $\sigma_k[W(R_k,U_k)-\min W]\to 1/4$.

The ordinary central path can select a different face point. For $F(n,u,v)=n^2/(1+u)$ on $n+u+v=1$, the secondary potential is $-(4\eta)^{-1}\log(1+u)$. Theorem 2 therefore selects $u_k\to 1$ for the original admissible fixed-step Burg map. In contrast, the unique minimizer of

$$F(n,u,v)-\mu(\log n+\log u+\log v), \qquad \mu\downarrow 0,$$

converges to a point with

$$u\longrightarrow\frac{\sqrt{41}-1}{10}, \qquad v\longrightarrow\frac{11-\sqrt{41}}{10}.$$

Supplement Section S4.10 proves this directly from the exact stationary equations and boundary capture. Thus the nonlinear Burg selection cannot be obtained by identifying its original orbit with the ordinary unweighted central path.

4 Reciprocal order, support actions, and nonunique selection

A fixed reciprocal-order anchor gives a complementary mechanism which needs neither an isolated secondary minimum nor a secondary rate. It permits nonlinear denominators and loss of maximal static support. The fixed-anchor condition need not hold for competing denominator directions, so it is a separate sufficient mechanism rather than a general nonunique-selection theorem.

Theorem 4 (Exact reciprocal-order length and support)**.** *Let $X_k = (u_k, x_{1,k}, \dots, x_{d,k})$ be any sequence in the open unit simplex. Put $r_k = 1/u_k$, $d_{0,k} = 0$, and $d_{i,k} = 1/x_{i,k} - 1/u_k$. If $\Delta d_{i,k} \geq 0$ for every $i \geq 1$ and $k \geq K$, then u_k is nondecreasing, X_k has finite physical length and converges to X_∞, with $u_\infty \geq u_K > 0$. For $M > k \geq K$,*

$$\sum_{l=k}^{M-1} \|\Delta X_l\|_1 \leq 2(r_k - r_M) = \frac{2(u_M - u_k)}{u_k u_M}. \tag{M.28}$$

The infinite tail satisfies the same bound with $M = \infty$. For an exact normalized Burg sequence with positive steps η_k, a sufficient condition is one fixed coordinate with

$$\partial_u F(X_k) \leq \partial_i F(X_k) \quad \text{for all } i, k \geq K. \tag{M.29}$$

Under this condition define the possibly infinite actions

$$B_i = \sum_{k \geq K} \eta_k (\partial_i F(X_k) - \partial_u F(X_k)), \qquad D_i = d_{i,K} + B_i.$$

Then $x_{i,\infty} > 0$ if and only if $B_i < \infty$, and for these coordinates

$$x_{i,\infty} = \frac{1}{r_\infty + D_i}, \qquad \frac{1}{r_\infty} + \sum_{i : B_i < \infty} \frac{1}{r_\infty + D_i} = 1. \tag{M.30}$$

The normalizer is unique on the interval of positive denominators. No optimization, convexity, analyticity, coordinate-ratio or rate assumption is needed for the geometric length assertion.

Proof. At each actual pair of endpoints set $c_i = \Delta d_i$, $c_0 = 0$, $\omega_i = x_i x_i^+ > 0$ (including $x_0 = u$), and $W = \sum_i \omega_i \leq 1$. The exact identity and mass conservation give

$$\Delta x_i = -\omega_i(\Delta r + c_i), \qquad \Delta r = -\frac{\sum_i \omega_i c_i}{W} \leq 0.$$

Writing $P = \sum_i \omega_i c_i = -W \Delta r$ yields

$$\|\Delta X\|_1 \leq P + W|\Delta r| = 2W(-\Delta r) \leq 2(-\Delta r).$$

Since $r \geq 1$, summation proves finite length, convergence, and (M.28). In the Burg case the common actual multiplier cancels exactly:

$$\Delta d_{i,k} = \eta_k(\partial_i F(X_k) - \partial_u F(X_k)).$$

Thus $d_i \to D_i$ in $\mathbb{R} \cup \{+\infty\}$, while r has a finite limit. The reciprocal formula gives precisely the support dichotomy and (M.30). Its left side is strictly decreasing in r where denominators are positive, proving uniqueness. The actions depend on the realized trajectory; this is not a static optimizer formula. ∎

The estimate also gives a chord–arc bound $2u_K^{-2}\|X_M - X_k\|_\infty$ for each segment's total ℓ^1 length. Non-anchor coordinates need not be monotone. Order depends only on the simplex derivative differences and is unchanged by adding a common scalar to the ambient gradient.

Proposition 4.1 (Summable violations of one fixed anchor)**.** For the same reciprocal chart allow $c_i = \Delta d_i$ of either sign. Put

$$N_k = \sum_i \omega_i(-c_i)_+, \qquad V_K = \sum_{k \geq K} \frac{N_k}{W_k}.$$

If $V_K < \infty$, then $u_k \geq (r_K + V_K)^{-1}$, the whole sequence converges with finite physical length, and

$$\sum_{k \geq K} \|\Delta X_k\|_1 \leq 2(r_K - r_\infty) + 4V_K \leq 2(r_K - 1) + 4V_K. \tag{M.31}$$

The same bound applies on every later tail. A sufficient condition is $\sum_{k \geq K} \sum_i (-\Delta d_{i,k})_+ < \infty$, or, for the exact Burg map,

$$\sum_{k \geq K} \eta_k \sum_{i \neq u} (\partial_u F(X_k) - \partial_i F(X_k))_+ < \infty.$$

Proof. Put $P_k = \sum_i \omega_i (c_i)_+$. Exact normalization gives $\Delta r = (N - P)/W$, so $(\Delta r)_+ \leq N/W$. Total positive variation of r is at most V_K and, because $r \geq 1$, total negative variation is at most $r_K - 1 + V_K$. Thus r has a finite limit and the stated upper bound. At each step,

$$\|\Delta X\|_1 \leq P + N + |P - N| = 2\max(P, N) \leq 2(-\Delta r)_+ + 2N.$$

Here $W \leq 1$ was used. Since $\sum N \leq V_K$ and $\sum(-\Delta r)_+ = r_K - r_\infty + \sum(\Delta r)_+$, summation proves (M.31). Also $N/W \leq \sum_i (-c_i)_+$, proving the sufficient condition. ∎

Under the unweighted sufficient condition each d_i has a finite or positive infinite limit; finiteness is equivalent to finite cumulative positive increments. This gives the analogous support criterion. The weighted budget alone is a sufficient geometric condition, not a necessary characterization of all finite-length orbits. A varying preferred coordinate is not covered without a budget for one fixed anchor.

Corollary 4.1 (Ordered positive-concave perspectives)**.** *On the simplex with normal coordinates n_j and tangential coordinates $x = (u, y_1, \dots, y_m)$, let*

$$F(n, x) = \sum_{j=1}^{q} \frac{n_j^{p_j}}{A_j(x)^{p_j - 1}}, \qquad p_j \geq 2 \text{ even.}$$

Suppose the denominators are positive concave real analytic on an open convex neighborhood of the full projected simplex $x \geq 0$, $\sum x_i \leq 1$, and throughout that set

$$\partial_u A_j \geq 0, \qquad \partial_u A_j \geq \partial_{y_i} A_j \quad \text{for every } i, j. \tag{M.32}$$

Every interior-start orbit of (M.1) *converges to a minimizer and has finite Euclidean and ℓ^1 physical length, with nondecreasing u, bounds* (M.28)*, and support criterion* (M.30)*. This permits nonunique secondary minima and nonlinear denominators. Analyticity can be dropped if the same convex smooth setting and derivative inequalities hold.*

Proof. The perspective n^p/a^{p-1} is convex for signed n, positive a, and is nonincreasing in a. Concave composition gives convexity across the simplex boundary. Positivity gives analyticity and an ambient Hessian bound; the Burg proximal problem has a unique interior target. With $C_j = (p_j - 1)n_j^{p_j}/A_j^{p_j} \geq 0$,

$$\partial_u F = -\sum_j C_j \partial_u A_j \leq 0, \qquad \partial_{y_i} F - \partial_u F = \sum_j C_j(\partial_u A_j - \partial_{y_i} A_j) \geq 0,$$

and $\partial_{n_j} F = p_j n_j^{p_j - 1}/A_j^{p_j - 1} > 0$ in the interior. Theorem 4 proves finite length and convergence before any rate or selection argument.

To identify the limit as minimizing, the usual convex comparator and sufficient decrease at the original step give, for every fixed interior z,

$$\eta[F(X_{k+1}) - F(z)] \leq D_h(z, X_k) - D_h(z, X_{k+1}), \qquad \eta k[F(X_k) - F(z)] \leq D_h(z, X_0).$$

For any minimizer X_* choose the interior $z_k = (1 - 1/k)X_* + X_0/k$. Bounded derivatives give $F(z_k) - \min F = O(1/k)$ and the literal Burg formula gives $D_h(z_k, X_0) = O(\log k)$. Thus $F(X_k) - \min F = O(\log k/k) \to 0$, without a boundary comparator or uniqueness. Continuity identifies the already selected limit. Here $\min F = 0$, so all normal coordinates vanish. ∎

The derivative condition is a checkable global sufficient condition. Weighted aggregate inequalities along a particular orbit, or the violation budget of Proposition 4.1, can replace it only as explicit orbitwise hypotheses. Finite length is rate-free and therefore consistent with the arbitrarily slow finite cascades in Section 4.1.

4.1 Sharp cascades of arbitrary finite logarithmic depth

Successive nonlinear denominators can postpone point selection to any prescribed finite iterated-logarithmic scale. The family below has a nonunique minimizing face, yet

every interior initialization selects the same vertex. Its coordinate derivatives have a fixed order, so Theorem 4 already supplies rate-free finite physical length. The reciprocal clocks give the selected point, every leading constant, and the sharp length tail.

Theorem 5 (A finite cascade with an exact physical tail)**.** *Fix an integer $m \geq 0$ and use the simplex coordinates*

$$X = (x_0, \dots, x_m, u, v), \qquad x_0 + \cdots + x_m + u + v = 1.$$

Define recursively

$$A_m = 1 + u, \qquad A_j = 2 - \frac{x_{j+1}^2}{A_{j+1}} \quad (0 \leq j < m), \qquad F(X) = \frac{x_0^2}{A_0}. \tag{M.33}$$

The function F is rational real analytic and convex on an open convex neighborhood of the entire closed simplex. Let $L > 0$ bound its ambient Hessian norm on that simplex. For every interior initialization and every fixed step $0 < \eta L < 1$, the exact normalized Burg iteration (M.1) *satisfies*

$$X_k \longrightarrow (0, \dots, 0, 1, 0), \qquad k x_{0,k}^2 \longrightarrow \frac{1}{2\eta}, \tag{M.34}$$

$$\ell_j(k) x_{j,k}^2 \longrightarrow 4 \quad (1 \leq j \leq m), \qquad \ell_{m+1}(k) v_k \longrightarrow 8, \tag{M.35}$$

where $\ell_1(k) = \log k$ and $\ell_{j+1}(k) = \log \ell_j(k)$ are evaluated sufficiently late. All x_j and v eventually decrease, and u eventually increases. The actual Euclidean path-length tail has the sharp equivalent

$$\mathcal{L}_k = \sum_{l \geq k} \|X_{l+1} - X_l\|_2 \sim \frac{8\sqrt{2}}{\ell_{m+1}(k)}. \tag{M.36}$$

For $m = 0$ the intermediate list in (M.35) *is empty and the tail in* (M.36) *is $8\sqrt{2}/\log k$.*

To see the connection with Section 4, put

$$P_0 = 1, \qquad P_j = \prod_{i=0}^{j-1} \frac{x_i^2}{A_i^2} \quad (1 \leq j \leq m+1).$$

On the closed simplex, $1 \leq A_j \leq 2$ and

$$F_{x_j} = \frac{2P_j x_j}{A_j}, \qquad F_u = -P_{m+1}, \qquad F_v = 0.$$

For $j < m$ the identity

$$F_{x_{j+1}} = F_{x_j} \frac{x_j x_{j+1}}{A_j A_{j+1}}$$

therefore gives the fixed derivative order $F_u \leq F_v \leq F_{x_m} \leq \cdots \leq F_{x_0}$. The sharper conclusions require more than this order. Supplement Section S5 proves them directly from the exact reciprocal differences. It first establishes all coordinate scales up to two-sided

constants, then selects the vertex, and only afterward determines the limiting denominators and sharp constants. The final step uses the genuine source–target normalization to prove eventual coordinate monotonicity and the path-length equivalent.

At depth $m = 1$, the objective is $F = x_0^2/(2 - x_1^2/(1+u))$. On the primary minimizing face $x_0 = 0$, the secondary maximum set of A_0 is $\{x_1 = 0,\ u + v = 1\}$. Its maximal support is $\{u, v\}$, whereas the selected vertex has $v = 0$. Thus finite physical length is compatible with loss of maximal secondary support, already at this depth.

Across the increasing-dimensional family, no single finite logarithmic depth supplies a universal tail scale: for every fixed $r \geq 1$ and $\alpha > 0$, any $m \geq r$ gives $\mathcal{L}_k \neq O(\ell_r(k)^{-\alpha})$. Each individual orbit nevertheless has finite physical length. The theorem asserts neither arbitrary depth in a fixed dimension nor a classification of general nonunique convex analytic objectives.

5 Scope and conclusions

The general-clock theorem separates a verifiable residual budget from the geometry of a unique analytic minimum. It permits arbitrary boundary gap patterns and supported degeneracy, but requires actual positive-free capture and ordinary dual errors of order $\beta T^{-3/2}$. Merely summable errors, uncaptured free variables, nonvanishing effective steps, and nonanalytic objectives are not covered. Its calibrated profile concerns the constructed adapted chart; it supplies neither an unweighted bounded-variation theorem for the further scaled variables nor a universal physical tail equivalent.

The first perspective reduction requires a unique secondary minimum. The second covers a nonunique secondary face only under full projected flatness and a uniformly positive transverse quadratic form, followed by a unique tertiary minimum. It allows that tertiary point to lie on any face with zero gaps or a degenerate supported Hessian. Proper secondary slices without controlled moving-fiber elimination, singular transverse forms and nonunique tertiary minima remain outside its statement. Neither exact reduction assumes an instantaneous normal equilibrium.

The reciprocal-order theorem is complementary. It can handle nonunique nonlinear problems with a fixed derivative-order anchor, or a verified finite violation budget for that anchor, without computing any clock. It does not provide that budget when the preferred coordinate changes. Its support actions are orbit-dependent, not a static variational selection formula. The sharp cascade exhibits arbitrarily large finite logarithmic depth as dimension increases, while retaining finite physical length. It is not a fixed-dimension classification or an infinite-length example.

The complete proof dependency is therefore finite: inherited static flag and literal Gram algebra; an actual-clock inexact dynamical theorem; two verified exact perspective reductions; and independent reciprocal-order and cascade arguments for support and tail

scope. No fixed-step dynamical conclusion is transferred to a variable-step sequence by a formal change of time.

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